

Arjun Challa

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Education2

Urbana High School

Ijamsville, MD | Aug 2023 - May 2027

Relevant Coursework: Calculus AB-BC, Physics Mechanics, Microeconomics, Macroeconomics, Computer Science A, Statistics, Psychology, Literature, Computer Science Principles, Environmental Science

Frederick Community College (STEM A.S.)

Frederick, MD | Aug 2025 - May 2027

Relevant Coursework: Virtualization and Cloud Essentials, Cloud System Administration, Introduction to Matlab

Skills

Languages: Python, SQL, R, PHP, HTML/CSS, Matlab

Tools: Scikit-learn, TensorFlow, Flask, Pandas, PyTorch, OpenCV, Git

Experience

Lead AI Technical Intern @ MedStar-Georgetown Center for AI in Healthcare

January 2026 – April 2026

- Developed and led a Python-based FDA medical device data pipeline to extract, clean, and validate authorization records from 2013 to 2023 across 510(k), De Novo, and PMA pathways by integrating bulk JSON extraction and web scraping.
- Managed AI-enabled classification, data linkage, and code quality efforts to analyze post-authorization outcomes, supporting a reproducible dataset and technical report on commercialization patterns in medical devices.

Software Intern @ North Carolina State University

April 2025 – Oct 2025

- Developed TensorFlow-based flood prediction models using USGS, NOAA, soil, and geospatial data to identify high-risk flooding zones in Raleigh, NC.
- Collaborated with a team of Ph.D. researchers and undergraduate students to conduct hotspot detection and model evaluation across multiple watersheds, aligning predictions with FEMA claims and historical flood events.

Research Assistant @ American University: Department of Math & Statistics

June 2025 – August 2025

- Conducted research under Dr. Hamid Semiyari on mean-variance optimization using quadprog in R to analyze portfolio risk-return dynamics across different stock selections and market conditions (2020–2023).
- Made and presented research findings on allocation sensitivity, diversification effects, and market-driven portfolio behavior.

TSRL Data Lab Member @ University of Southern California

March 2025 – Nov 2025

- Conducted hands-on project experiences analyzing athlete data to build predictive performance models and applying Artificial Intelligence techniques to real-world training scenarios.

Projects & Awards

Simple Proctor | Python, Flask, OpenCV, MediaPipe, NumPy

March 2026

- Developed a computer vision-based exam proctoring system with a Flask web interface, streaming live video, and performing real-time face-count detection, session monitoring, and alert generation.
- Created a detection pipeline using MediaPipe with OpenCV fallback logic, cooldown violation handling, screenshot capture, and alert/stat tracking for monitoring reliability. View at https://github.com/achalla18/simple_proctor

WebML Resume | Python, Flask, scikit-learn

January 2026

- Built an ML resume-screening web application that parses uploaded PDFs, extracts text, and scores technical fit for ML/SWE roles using an NLP classification pipeline. View at <https://github.com/achalla18/webml-resume>

KalView | Python, Flask, JavaScript, GDELT API, USGS API, World Bank API

April 2026

- Developed an interactive intelligence dashboard that visualizes geopolitical signals through a live map interface, allowing users to explore real-world events by region. View live demo at <https://kal-view.vercel.app/> (Give it a minute)
- Integrated data sources such as USGS Earthquake API and World Bank indicators to display relevant news with article-based confidence scoring to help users understand the reliability of situations. View code at <https://github.com/achalla18/KalView>

Hi-Lite | Founder ... Used: React, Bootstrap, HTML/CSS, JavaScript, MySQL

May 2025 – Present

- Designing and developing a social media platform for high school students to showcase digital portfolios, feature projects, and receive skill endorsements.
- Developed a responsive UI/UX and currently implementing backend functionality to support user authentication, profile management, and project posting.

Agriimpact | Python, Flask, NumPy, HTML/CSS, USDA QuickStats API

April 2025

- Developed a web application enabling small and rural farmers to compare crop profitability, risk, and government support using real USDA data.
- Implemented Monte Carlo simulations (1,000 runs) to assess yield and price variability, and presented results in a simple, interactive interface to support informed decision-making. View at <https://github.com/achalla18/agri-impact>